WEEK 1: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub-Strand: Conserving Animal Feed: Hay

Specific Learning Outcomes:
By the end of the lesson, learners will be able to:
- State the meaning of forage as a method of conserving animal feed.
- Discuss the types of materials used in making forage in the environment.
- Search for information on the materials used in making forage in the environment.
- Appreciate the materials used in making forage in the environment.

Key Inquiry Question(s):
- What is forage?
- Which materials are used in making forage in the environment?

Learning Resources:
- Lesson notes
- Digital devices
- Agriculture & Nutrition Learner's Textbook
- Pictures

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on resource conservation.
- Guide learners to read and discuss relevant content from the learning resources, focusing on the definitions and importance of forage.

Lesson Development (30 minutes)

Step 1: Understanding Forage (10 minutes)
- Activity: In pairs, learners discuss and develop a dictionary definition of "forage."
- Outcome: Learners share their definitions, highlighting the key elements related to conserving animal feed.

Step 2: Identifying Materials (10 minutes)
- Activity: Brainstorm materials found in the local environment suitable for making forage, such as grasses, legumes, and hay.

- Outcome: Each pair presents their findings, and the teacher compiles a comprehensive list on the board.

Step 3: Researching Forage Materials (5 minutes)
- Activity: Learners use digital devices or textbooks to research one specific material that can be used to make forage.
- Outcome: Learners take notes on their findings, focusing on properties, benefits, and methods of preparation.

Step 4: Discussion and Presentation (5 minutes)
- Activity: Groups share their research insights with the class.
- Outcome: Through guided discussion, learners appreciate the variety of materials and their roles in making effective forage.

Conclusion (5 minutes)
- Summary: Recap the definition of forage, the materials used, and their importance in animal feed conservation.
- Interactive Activity: Quick quiz or application game where learners name materials used in forage.
- Preview Next Session: Provide a teaser for the next topic, possibly discussing the nutritional benefits of various forages.

Extended Activities:
- Forage Material Exploration: Assign learners to collect a sample of a local forage material (or a picture) and prepare a short presentation on its benefits and uses for animal feed.
- Field Trip: Plan a visit to a local farm or a forage production facility, allowing students to see the practical applications of the lessons learned.
- Create a Forage Recipe Book: Students collaborate to compile a recipe book featuring different forage mixes that can be used for various livestock, including care tips related to their preparation.

Teacher Self-Evaluation:

WEEK 1: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Animal Feed: Hay

Specific Learning Outcomes:
By the end of the lesson, learners will be able to:
- Outline the effects of drought on livestock feed resources.
- Discuss the effects of drought on livestock feed resources.
- Search for information on the effects of drought on livestock feed resources.
- Acknowledge the effects of drought on livestock feed resources.

Key Inquiry Question:
- What are the effects of drought on livestock feed resources?

Learning Resources:
- Digital devices (tablets/laptops)
- Lesson notes
- Agriculture and Nutrition Learner's Textbook

Organization of Learning:

Introduction (5 minutes)
1. Review the previous lesson on animal feed resources and their importance for livestock.
2. Introduce the key inquiry question: “What are the effects of drought on livestock feed resources?”
3. Guide learners in reading relevant sections of the Agriculture and Nutrition Learner's Textbook, focusing on drought impacts.

Lesson Development (30 minutes)

Step 1: Brainstorming Session (10 minutes)
- In small groups, learners brainstorm and write down their initial thoughts on how drought might affect livestock feed resources.
- Each group shares one point with the class to kickstart a discussion.

Step 2: Research (10 minutes)
- Using digital devices, learners search for information on how drought affects livestock feed. Encourage the use of trusted websites or agricultural databases.

- Groups compile notes from their findings focusing on key aspects such as potential shortages, nutritional impacts, and economic consequences.

Step 3: Group Discussion (5 minutes)
- Groups discuss amongst themselves the information they found, comparing and contrasting the effects of drought based on their research.
- A representative from each group prepares to present their findings to the class.

Step 4: Class Presentations (5 minutes)
- Groups present their findings to the class. Each presentation should highlight at least three significant effects of drought on livestock feed resources.
- Encourage questions and discussions after each presentation to enhance understanding.

Conclusion (5 minutes)
1. Summarize the key points discussed during the lesson, emphasizing the impacts of drought on livestock feed resources.
2. Conduct a brief interactive activity, such as a quick quiz or audience poll, to reinforce the main topics covered.
3. Preview the next session, which will focus on strategies for conserving resources during drought conditions, and pose questions for learners to consider.

Extended Activities:
1. Research Project: Assign a project where learners choose a specific region affected by drought and create a report or presentation on its effects on local livestock feed resources.
2. Field Trip: Plan a visit to a local farm or agriculture extension office to observe real-world impacts of drought on livestock operations.
3. Guest Speaker: Invite a local farmer or agricultural expert to discuss their experiences with drought and strategies they use to manage livestock feed resources during dry conditions.

Teacher Self-Evaluation:

WEEK 1: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Animal Feed: Hay

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Identify the methods of conserving forage in coping with drought.
2. Describe baled haymaking as a method of conserving forage in coping with drought.
3. Search the internet or textbook for information on the process of preparing baled haymaking as a method of conserving forage.
4. Acknowledge the process of preparing baled haymaking as a method of conserving forage.

Key Inquiry Question:
- How can hay conservation contribute to coping with drought?

Learning Resources:
- Digital devices (laptops/tablets)
- Agriculture and Nutrition Learner's Textbook
- Video clips demonstrating baled haymaking

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson, discussing the importance of sustainable practices in agriculture.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing key concepts related to drought and forage conservation.

Lesson Development (30 minutes)
- **Step 1:** Identify Methods of Conserving Forage
- In small groups, students brainstorm various methods used to conserve forage during drought conditions.
- Each group writes down their findings to share with the class.

- **Step 2:** Research Baled Haymaking
- Students use digital devices to research baled haymaking, focusing on its benefits and challenges as a method of conserving forage.
- They should find at least three key points to discuss later.

- **Step 3:** Explore Plant Materials

- Each group identifies different types of plant materials commonly used in making baled hay.
- They prepare a list of these materials and share with the class, discussing their availability and suitability for baled haymaking.

- **Step 4:** Outline the Baled Haymaking Process
- Groups outline the process of preparing baled hay, noting each step involved.
- They will present this outline to the class along with any specific tools or techniques needed.

Conclusion (5 minutes)
- Summarize key points discussed during the lesson—methods of conserving forage and the process of baled haymaking.
- Conduct an interactive quiz or discussion to reinforce the main topics learned in the lesson.
- Prepare learners for the next session by previewing upcoming topics such as "Impact of Drought on Livestock" or questions to consider like, "What other methods could be used to conserve livestock feed?"

Extended Activities:
- Research Assignment: Students can conduct a detailed report on different haymaking techniques around the world and how they adapt to various climates.
- Field Trip: Organize a visit to a local farm to observe the baled haymaking process first-hand and discuss challenges faced by farmers during drought conditions.
- Creative Project: Have students create a poster or presentation on the best practices for conserving forage, highlighting the importance of sustainable agriculture methods.

Teacher Self-Evaluation:

WEEK 1: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Animal Feed: Hay

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify the types of bales used in baled haymaking.
- Discuss the storage methods of baled hay in coping with drought.
- Search for clips and information on the storage methods of baled hay.
- Acknowledge the need for proper storage of baled hay in coping with drought.

Key Inquiry Questions:
- Why is it important to store baled hay properly?
- How is baled hay stored in order to cope with drought?

Learning Resources:
- Lesson notes
- Digital devices
- Agriculture and Nutrition Learner's Textbook
- Pictures of hay bales
- Video clips demonstrating hay storage methods

Organisation of Learning:

Introduction (5 minutes)
1. Begin with a quick review of the previous lesson on the basics of haymaking.
2. Ask students to share what they know about why hay is an important feed for animals, leading into the discussion of storage.
3. Guide learners to read relevant content from their textbooks, focusing on the importance of proper hay storage.

Lesson Development (30 minutes)

Step 1: Identify Types of Bales (10 minutes)
- Present various pictures of hay bales (square, round, etc.) around the classroom.
- In small groups, students discuss the different types of bales they see.
- Each group shares which type of bale might be most beneficial for different farming scenarios.

Step 2: Discuss Storage Methods (10 minutes)

- Lead a discussion on the importance of storing hay properly, especially in drought conditions.
- Facilitate groups using digital devices to research different storage methods (e.g., barns, tarps, humidity control) and their benefits in preventing hay spoilage.

Step 3: Share Research Findings (5 minutes)
- Have each group present their findings on one storage method, highlighting its pros and cons.
- Encourage students to think critically about how these methods can help during drought.

Step 4: Reinforce Importance of Proper Storage (5 minutes)
- Ask students to consider and discuss as a class why proper storage is crucial for farmers facing drought.
- Students should connect this importance back to sustainability and animal welfare.

Conclusion (5 minutes)
- Summarize the key points discussed during the lesson, reiterating the types of bales and storage methods.
- Conduct a brief interactive quiz or poll asking students to vote on the most critical aspect of hay storage using a digital tool (Kahoot, Mentimeter, etc.).
- Preview the next session's topic on alternative animal feeds and encourage students to think about other factors affecting animal nutrition.

Extended Activities:
1. Field Trip: Organize a visit to a local farm to observe hay storage methods in action.
2. Project: Have students create a presentation or poster on a specific storage method, including best practices for maintaining hay quality during drought.
3. Research Assignment: Students can choose a country's hay production and storage methods to compare with their own, focusing on how different climates impact the practice.

Teacher Self-Evaluation:

WEEK 2: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub-Strand: Conserving Animal Feeds

Specific Learning Outcomes:
By the end of the lesson, students should be able to:
- State the meaning of standing forage as a method of conserving forage.
- Describe the preparation and management of standing forage to cope with drought.
- Use digital or print resources to search for information on standing forage as a method of conserving animal feeds.
- Acknowledge the benefits of standing forage as a method of conserving animal feeds.

Key Inquiry Question:
- What are the benefits of leaving forage standing in the field?

Learning Resources:
- Lesson notes
- Digital devices
- Agriculture and Nutrition Learner's Textbook
- Pictures of standing forage

Organisation of Learning:

Introduction (5 minutes)
- Briefly review the content of the previous lesson regarding animal feeds.
- Ask students to share anything they remember about methods of conserving animal feeds.
- Facilitate group reading of relevant sections in the textbook, emphasizing the definition and significance of standing forage.

Lesson Development (30 minutes)
- **Step 1:** Group Discussion
- In groups, students brainstorm what they think standing forage means.
- Prompt students to think about their prior knowledge or experiences related to grazing and feed conservation.

- **Step 2:** Research
- Guide students to use both digital devices and textbooks to find information on how standing forage helps conserve animal feed.
- Encourage them to focus on key factors such as benefits, preparation, and management.

- **Step 3:** Sharing Experiences
- Have each group present what they found in their research.
- Facilitate a class discussion on how different regions or types of farms may utilize standing forage differently.

- **Step 4:** Preparation and Management
- Discuss the methods for preparing standing forage, such as the timing of mowing, and how to manage it through drought conditions.
- Highlight the importance of strategic planning for livestock feeding during droughts.

Conclusion (5 minutes)
- Summarize the key points learned about standing forage and its benefits for animal feeding.
- Conduct a quick interactive activity such as a quiz or a think-pair-share to reinforce the main topics discussed.
- Give a brief preview of the next lesson, suggesting they think about other methods of feed conservation for discussion.

Extended Activities:
- Field Trip: Organize a visit to a local farm that uses standing forage to observe the practice firsthand.
- Research Project: Assign a project where students can choose a specific type of forage and report on its benefits, uses, and challenges associated with using it as animal feed.
- Creative Media: Have students create a poster or digital presentation that highlights the importance of standing forage in agricultural sustainability.

Teacher Self-Evaluation:

WEEK 2: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Animal Feeds

Specific Learning Outcomes:
By the end of the lesson, students should be able to:
- Outline the techniques for managing standing forage in coping with drought.
- Discuss the techniques for managing standing forage in coping with drought.
- Search the internet for information on the ways of managing standing forage to cope with drought.
- Acknowledge the ways of managing standing forage effectively.

Key Inquiry Question:
- How can we manage standing forage in order to cope with drought?

Learning Resources:
- Lesson notes
- Digital devices (tablets/laptops)
- Agriculture and Nutrition Learner's Textbook

Organisation of Learning

Introduction (5 minutes)
- Review the previous lesson on the importance of forage and its role in animal nutrition.
- Guide learners to read and discuss relevant content from the Agriculture and Nutrition Learner's Textbook, emphasizing key concepts about standing forage and its significance, especially during drought conditions.

Lesson Development (30 minutes)
Step 1: Brainstorming Session (10 minutes)
- Divide students into small groups and ask them to brainstorm possible techniques for managing standing forage to cope with drought. Encourage them to think about traditional and modern methods.
- Each group writes their ideas on a shared document or whiteboard.

Step 2: Research Activity (10 minutes)
- In their groups, students use digital devices to search for additional information about managing standing forage during drought. They should look for at least three specific strategies or techniques.

- Encourage them to focus on practical approaches like rotational grazing, moisture retention practices, and selecting drought-resistant forage crops.

Step 3: Group Discussion (5 minutes)
- Groups come together and share their findings. Each group presents their researched techniques, explaining how each method supports standing forage management during drought.
- As groups present, encourage other students to ask questions and engage in discussion.

Step 4: Summarizing Techniques (5 minutes)
- As a class, compile a list of the techniques discussed by each group on the board. Highlight the most effective strategies and encourage students to add any additional thoughts.

Conclusion (5 minutes)
- Summarize the key points covered in the lesson, reiterating the importance of managing standing forage to cope with drought.
- Conduct a brief interactive quiz or activity (e.g., a Kahoot!) to reinforce the main topics.
- Preview the next lesson which will focus on the implications of drought on livestock health and productivity, prompting students to think about ways to maintain animal welfare during drought conditions.

Extended Activities:
1. Field Study: Organize a visit to a local farm or botanical garden where students can observe different types of forage crops and their management strategies during drought.
2. Research Project: Assign students to choose a particular forage management technique and create a presentation or report detailing how it helps in drought scenarios.
3. Guest Speaker: Invite a local agricultural expert to provide insight into real-world challenges and solutions related to forage management and drought resilience.

Teacher Self-Evaluation:

WEEK 2: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Animal Feeds: Hay

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- State the meaning of stacking as a method of conserving forage.
- Describe stacking as a method of conserving forage.
- Search the internet or textbook for information on stacking as a method of conserving forage.
- Adopt conservation of forage in coping with drought.

Key Inquiry Questions:
- What is stacking?
- How is forage prepared by stacking stored?

Learning Resources:
- Lesson notes
- Digital devices (computers, tablets)
- Agriculture and Nutrition Learner's Textbook
- Video clips about stacking
- Pictures illustrating the stacking process

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on forage and its importance in animal nutrition.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing the concepts of stacking and its necessity in conserving forage.

Lesson Development (30 minutes)

Step 1: Define Stacking (10 minutes)
- In pairs, learners will discuss the definition of stacking as a conservation method.
- Each pair will then share their definition with the class, ensuring all students understand the basic concept.

Step 2: Research Stacking Methods (10 minutes)

- Learners will use digital devices or textbooks to search for detailed information on how stacking is done.
- Encourage them to focus on different methods, materials used, and the benefits of stacking for animal feed conservation.

Step 3: Group Discussion (5 minutes)
- In small groups, learners will discuss their findings, comparing different stacking methods they discovered during their research.
- They should also talk about the impact of these methods on conserving resources during droughts.

Step 4: Class Sharing (5 minutes)
- Each group will present a summary of their discussion to the class.
- The teacher will provide feedback and clarify any misconceptions about stacking and its role in forage conservation.

Conclusion (5 minutes)
- Summarize the key points discussed: the meaning of stacking, methods, and benefits for coping with drought.
- Conduct a brief interactive activity, such as a quick quiz or a word cloud on what they learned about stacking.
- Set the stage for the next session, which will focus on practical applications of stacking in local farms. Pose questions for them to consider for the next class.

Extended Activities:
- Field Visit: Arrange a visit to a local farm where stacking is practiced. Students can observe firsthand and ask questions.
- Create a Poster: Students can create informative posters about stacking methods and display them around the classroom.
- Drought Simulation: Conduct a classroom activity where students simulate drought conditions and plan which forage conservation methods (like stacking) they would implement.

Teacher Self-Evaluation:

WEEK 2: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Animal Feeds

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. State the meaning of box-bailing as a method of conserving forage.
2. Describe how box-bailing is done as a method of conserving forage.
3. Use digital devices to search for information and clips on box-bailing as a method of conserving forage.
4. Adopt box-bailing as a conservation method of forage in coping with drought.

Key Inquiry Questions:
1. What are the benefits of box-bailing as a method of conserving forage?
2. How is box-bailing done as a method of conserving forage?

Learning Resources:
- Lesson notes
- Digital devices (tablets/laptops/smartphones)
- Internet access
- Video clips on box-bailing
- Agriculture and Nutrition Learner's Textbook

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on forage and its importance in animal nutrition.
- Ask students to share what they remember and prompt a discussion about the challenges in conserving animal feeds, leading into the topic of box-bailing.

Lesson Development (30 minutes)

Step 1: Understanding Box-Bailing (10 minutes)
- In small groups, learners brainstorm and present their ideas on what box-bailing means. Guide them to consider why it may be important for conserving forage.
- Each group shares their definitions, and the teacher clarifies any misconceptions.

Step 2: Researching Box-Bailing (10 minutes)
- Students will use digital devices to search for information and find clips on how box-bailing is conducted.

- Suggest specific websites or keywords to use, and remind them to note down important points and sources.

Step 3: Discussing the Process (5 minutes)
- Bring the class back together and have each group present how box-bailing is done. Encourage learners to include steps, tools needed, and any methods discussed in the videos they reviewed.
- Teacher provides additional information to deepen understanding.

Step 4: Benefits and Adoption of Box-Bailing (5 minutes)
- Engage students in a discussion on the benefits of using box-bailing, especially during drought conditions.
- Ask students how they might implement this method in their own agricultural practices or suggest ways it could be improved.

Conclusion (5 minutes)
- Summarize the key points learned, including the process of box-bailing and its benefits.
- Conduct a brief interactive activity: ask students to create a quick “top 3 tips” list on box-bailing, which they can share with the class.
- Preview the next lesson by hinting at discussing alternative forage conservation methods to further round their understanding.

Extended Activities:
1. Research Project: Have students investigate a specific type of forage and present on the best conservation methods, including box-bailing.
2. Field Trip: If feasible, arrange a visit to a local farm to observe or participate in actual box-bailing procedures.
3. Creative Assignment: Students can create a short video or presentation explaining box-bailing for a younger audience or community members.

Teacher Self-Evaluation:

WEEK 3: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Animal Feeds

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Conserve forage through the stacking method using locally available materials.
2. Enjoy preparing and conserving forage using the stacking method.

Key Inquiry Questions:
- How can hay conservation contribute to coping with drought?

Learning Resources:
- Grass and maize stovers
- School farm (or designated site for practical work)
- Digital devices (for research purposes)

Organisation of Learning:

Introduction (5 minutes)
- Review: Begin by reviewing the key points discussed in the previous lesson about the importance of resource conservation in agriculture.
- Read & Discuss: Invite learners to read short excerpts on forage conservation from digital devices. Discuss the role of hay in animal feed and drought resilience.

Lesson Development (30 minutes)

Step 1: Collecting Materials (10 minutes)
- Group Work: Divide the class into small groups. Assign each group the task of collecting locally available materials such as grass and maize stovers from the school farm or designated area.
- Observation: Encourage students to note the types of grass and stover that are best suited for hay production.

Step 2: Preparing Forage (10 minutes)
- Demonstration: Demonstrate how to prepare the collected materials. Show the best practices for cutting and arranging grass and stovers for effective hay making.
- Hands-On Practice: Allow groups to practice this technique, ensuring they understand how to handle the materials safely.

Step 3: Stacking the Forage (10 minutes)
- Instruction: Teach students about the stacking method for hay conservation. Explain how proper stacking can prevent spoilage and save feed for dry seasons.
- Group Activity: Groups will create a small stack of hay using the materials they prepared, ensuring they follow the guidelines discussed.

Step 4: Storing the Forage (Optional Extension)
- Briefly discuss the importance of proper storage for hay. If time allows, show students how to cover and secure their hay stack to protect it from weather elements.

Conclusion (5 minutes)
- Summary: Highlight the key points learned during the lesson: the importance of conserving forage, the stacking method, and how these practices contribute to drought resilience.
- Interactive Activity: Conduct a quick quiz or share interesting facts to reinforce learning. Encourage students to share their thoughts on what they learned.
- Preview Next Session: Inform students about the next lesson on alternative animal feeds during feed shortages, encouraging them to think about what feeds are available in their local environment.

Extended Activities:
- Research Project: Assign students to research different methods of forage conservation used worldwide and present their findings in the next class.
- Field Trip: If possible, arrange a field trip to a local farm that practices advanced forage conservation techniques to observe and learn directly from practitioners.

Teacher Self-Evaluation:

WEEK 3: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Animal Feeds

Specific Learning Outcomes:
- By the end of the lesson, learners should be able to conserve forage through the box-bailing method using locally available materials to make hay for the drought season.
- Learners should enjoy preparing and conserving forage using the box-bailing method.

Key Inquiry Question(s):
- How can we prepare hay using the box-bailing method?

Learning Resources:
- Strings
- Wooden boxes
- Environment (grassy areas)
- Grass and maize stovers
- Digital devices (for research or video demonstration)

Organisation of Learning:

Introduction (5 minutes):
- Begin with a quick review of the previous lesson on animal feeds and their importance.
- Pose the key inquiry question to the class: "How can we prepare hay using the box-bailing method?"
- Encourage quick brainstorming from learners to trigger their thinking about hay preparation.

Lesson Development (30 minutes):
- Organize learners into small groups and guide them through the following steps:

Step 1: Collecting Materials
- Activity: Guide groups to collect grass and maize stovers from the environment.
- Goal: Ensure that all groups have enough material for hay preparation.

Step 2: Constructing the Wooden Box
- Activity: Using locally available materials, each group constructs a wooden box that will be used to make hay.
- Goal: To understand the importance of proper structure in conserving forage.

Step 3: Preparing Forage
- Activity: Once the box is constructed, groups work collaboratively to prepare the forage by tightly packing the harvested materials into the box.
- Goal: To experience the process of box-bailing and learn how it helps conserve feed.

Step 4: Storing Hay Safely
- Activity: Discuss methods of storing the prepared hay to ensure it remains fresh and usable during drought season.
- Goal: To learn about the safe storage of forage to maximize its shelf life.

Conclusion (5 minutes):
- Summarize the key points covered during the lesson, reiterating the importance of the box-bailing method for conserving animal feeds.
- Conduct a brief interactive quiz or discussion to reinforce the main topics covered, such as identifying materials and steps needed in the box-bailing process.
- Prepare learners for the next session by previewing upcoming topics, such as the nutritional value of stored feeds.

Extended Activities:
- Home Project: Ask students to create a poster illustrating the box-bailing process, including materials needed and steps involved.
- Field Trip: Organize a visit to a local farm to see hay preparation in action.
- Research Assignment: Assign learners to research different hay preservation methods used around the world and present findings to the class.

Teacher Self-Evaluation:

WEEK 3: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Animal Feeds

Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Outline the strategies households can adopt in conservation of forage to cope with drought.
2. Discuss the strategies households can adopt in conservation of forage to cope with drought.
3. Adopt conservation practices for forage in coping with drought.

Key Inquiry Question:
- How can households adopt conservation of forage in coping with drought?

Learning Resources:
- Lesson notes
- Digital devices
- Agriculture and Nutrition Learner's Textbook

Organisation of Learning:

Introduction (5 minutes)
- Begin with a quick review of the previous lesson, asking students to share key points.
- Introduce today's topic on conserving animal feeds, focusing on the impact of drought on forage availability.
- Distribute learning resources and guide learners to read a section that relates to drought and forage conservation.

Lesson Development (30 minutes)

Step 1: Brainstorming Session (10 minutes)
- In groups, learners brainstorm ideas about the importance of conserving forage and how drought affects animal feed.
- Each group writes down their initial thoughts on a shared digital document.

Step 2: Research (10 minutes)
- Using digital resources, learners research effective strategies for conserving forage during drought periods.
- Encourage them to focus on both traditional and modern methods.

Step 3: Discussion (5 minutes)

- Groups discuss their findings and come up with a comprehensive list of strategies.
- Facilitate a conversation about the pros and cons of each strategy.

Step 4: Class Presentation (5 minutes)
- Each group presents their findings to the class, explaining one or two strategies in detail.
- Encourage peer feedback and questions after each presentation to foster engagement.

Conclusion (5 minutes)
- Summarize key points from the presentations and the overall lesson on forage conservation.
- Reinforce the learning objectives achieved during the lesson through a brief review.
- Conduct a quick interactive activity, such as a quiz or brainstorming session on future drought preparation strategies.
- Provide a brief preview of the next session, introducing potential topics or questions related to sustainable farming practices.

Extended Activities:
1. Forage Management Plan: Students can create a plan for their own household or a fictional scenario outlining how they would manage forage conservation in the event of a drought.
2. Field Research: Encourage students to interview local farmers or agricultural experts about their experiences and strategies for forage conservation.
3. Community Project: Organize a school-based project where learners can implement one strategy of forage conservation in collaboration with a local farm or community garden.

Teacher Self-Evaluation:

WEEK 3: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Animal Feeds

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- State the benefits of hay as a method of conserving animal feeds.
- Discuss the benefits of hay in coping with drought.
- Create posters illustrating the benefits of hay as a method of conserving animal feeds.
- Acknowledge the importance of hay in feed conservation.

Key Inquiry Question(s):
- What are the benefits of hay as a method of conserving animal feeds?

Learning Resources:
- Lesson notes
- Digital devices (computers/tablets)
- Posters
- Agriculture and Nutrition Learner's Textbook

Organisation of Learning:

Introduction (5 minutes)
- Begin by briefly reviewing the previous lesson on animal nutrition and feed types.
- Introduce the topic of hay as a conservation method by asking students what they know about hay and its use in agriculture.
- Provide an overview of the lesson objectives and activities.

Lesson Development (30 minutes)

Step 1: Brainstorm (10 minutes)
- Divide the class into small groups.
- Instruct each group to brainstorm and list the benefits of using hay as feed conservation. Encourage them to think about different scenarios, especially drought conditions.
- After brainstorming, groups will share their ideas with the class.

Step 2: Discussion (10 minutes)
- Each group will select a spokesperson to lead a discussion on their findings.

- Facilitate a class discussion about how hay can help save costs and ensure livestock health during times of feed scarcity, highlighting specific benefits like nutritional value and long storage life.

Step 3: Poster Creation (5 minutes)
- Prompt students to summarize the benefits they discussed on large poster sheets.
- They should include visuals and key points to make their posters engaging. Provide materials and digital resources if available.

Step 4: Display and Share (5 minutes)
- Allow each group to present their poster to the class, briefly explaining the benefits they highlighted.
- Hang their posters around the classroom as a visual reminder of what they learned.

Conclusion (5 minutes)
- Summarize the key points discussed about the benefits of hay in conserving animal feeds. Emphasize its role in managing feed scarcity during drought conditions.
- Conduct an interactive quiz or a flashcard game to reinforce the learning objectives.
- Preview the next lesson where students will learn about other feed conservation methods, prompting them to think about how they can implement these in real-life scenarios.

Extended Activities:
- Research Project: Assign students to research different types of hay and their specific nutritional benefits for various livestock. Present findings in the next class.
- Field Trip: Organize a visit to a local farm that uses hay for feed conservation to see firsthand its application and benefits.
- Creative Writing: Encourage students to write a short essay on "The Role of Hay in Sustainable Farming."

Teacher Self-Evaluation:

WEEK 4: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Leftover Food

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- State the importance of conserving leftover food at home.
- Explain the significance of conserving leftover food at home.
- Search the internet or print resources for information on the importance of conserving leftover food at home.
- Acknowledge the benefits of conserving leftover food at home.

Key Inquiry Questions:
- What is leftover food?
- What is the importance of conserving leftover food at home?

Learning Resources:
- Lesson notes
- Digital devices (tablets/laptops)
- Agriculture and Nutrition Learner's Textbook
- Posters or charts

Organisation of Learning

Introduction (5 minutes)
1. Review Previous Lesson: Ask students to share one thing they remember from the last class related to food conservation or resource management.
2. Discussion: Guide learners to read selected excerpts from the learning resources. Focus the discussion on understanding leftover food and its significance, encouraging students to share initial thoughts.

Lesson Development (30 minutes)

Step 1: Define Leftover Food (15 minutes)
- Activity: In small groups, learners will brainstorm what they understand by the term "leftover food."
- Guidance: Encourage them to think about different types of leftovers, such as cooked meals, raw ingredients that are not used, or food scraps.

- Presentation: Each group will share their definitions with the class. Write key points on the board for visual reinforcement.

Step 2: Share Personal Experiences (5 minutes)
- Activity: Learners share personal stories or experiences related to leftover food at home.
- Discussion Points: Focus on their practices on how leftovers are treated at home, whether they are discarded, reused, or shared.

Step 3: Research Importance (5 minutes)
- Activity: Use digital devices to search for articles or information on the importance of conserving leftover food.
- Resources: Provide a list of reputable websites or guide them to look into their textbooks for relevant information.
- Presentation: Each group presents one or two insights they discovered during their research.

Step 4: Group Discussion on Importance (5 minutes)
- Guided Discussion: Engage the class in a discussion about the benefits of conserving leftover food, such as reducing waste, saving money, and environmental impact.
- Wrap-Up: Summarize insights shared by groups, ensuring to highlight common themes.

Conclusion (5 minutes)
- Summary: Recap the key points discussed during the lesson regarding the meaning and importance of conserving leftover food.
- Interactive Activity: Conduct a quick quiz or fun game (e.g., "Two Truths and a Lie" about leftover food facts) to reinforce learning and engage students.
- Preview: Briefly introduce the topic for the next class, sparking curiosity about future lessons in agriculture and resource conservation.

Extended Activities:
- Research Project: Have students choose a specific food item and create a presentation on how to properly store, reuse, or repurpose leftovers related to that food item.
- Cooking Challenge: Encourage students to create a recipe using only leftovers that they have at home and bring it to share with classmates.

Teacher Self-Evaluation:

WEEK 4: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Leftover Food

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify the methods of conserving leftover food to avoid wastage.
- Describe the methods of conserving leftover food at home to avoid wastage.
- Search the internet or print resources for methods of conserving leftover food at home to avoid wastage.
- Embrace methods of conserving leftover food to minimize wastage.

Key Inquiry Question(s):
- Which methods do you use to conserve leftover food at home to avoid wastage?

Learning Resources:
- Lesson notes
- Digital devices (laptops, tablets)
- Posters or charts
- Agriculture and Nutrition Learner's Textbook

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on the importance of resource conservation.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing the key concept of food wastage.

Lesson Development (30 minutes)

Step 1: Brainstorming Methods (10 minutes)
- In pairs, learners brainstorm various methods they or their families use to conserve leftover food at home.
- Encourage each pair to share at least one method with the class, writing ideas on the board to create a comprehensive list.

Step 2: Research Phase (10 minutes)
- Using digital devices or print resources, learners search for additional methods of conserving leftover food.

- Each group records at least three new methods they discover during this session.

Step 3: Discussion of Methods (5 minutes)
- Groups reconvene and discuss the new methods they found.
- Emphasize collaboration by allowing groups to share their new findings with the class and adding them to the list created earlier.

Step 4: Poster Preparation (5 minutes)
- Each group selects their favorite methods from the combined lists and prepares a poster illustrating these methods.
- Posters should include both visual elements and descriptions of each method to educate others on conserving food.

Conclusion (5 minutes)
- Summarize the key points about the importance of conserving food and different methods discussed during the lesson.
- Conduct a quick interactive quiz where learners can answer questions based on the methods learned.
- Prepare learners for the next session by introducing the topic of food preservation techniques and encouraging them to think about what they've learned.

Extended Activities:
- Food Diary Assignment: Students maintain a food diary for a week, noting any leftover food and how it was conserved or wasted. They can reflect on their habits and suggest improvements.
- Community Project: Partner with local shelters or food banks for a project where students can help collect, store, and manage food items effectively to reduce waste.

Teacher Self-Evaluation:

WEEK 4: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Leftover Food

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Select at least one method of conserving leftover food to avoid wastage.
- Use the selected method to conserve leftover food.
- Embrace the methods of conserving leftover food at home.

Key Inquiry Question:
- How can you conserve leftover food to avoid wastage?

Learning Resources:
- Leftover food
- Digital devices (tablets/laptops)
- Lesson notes (printed or digital)

Organisation of Learning:

Introduction (5 minutes)
1. Review Previous Lesson: Begin by briefly revisiting the key points from the last lesson on food waste and its impacts. Ask students to share their thoughts on what they remember.
2. Guide Discussion: Facilitate a discussion about the importance of food conservation. Allow students to explore their ideas about how leftover food can be managed at home.

Lesson Development (30 minutes)

Step 1: Understanding Methods (10 minutes)
- In small groups, students will brainstorm and research different methods for conserving leftover food (e.g., refrigeration, freezing, pickling, composting). Use digital devices to look up various techniques.
- Each group will select one method they find most appealing or practical.

Step 2: Practical Application (10 minutes)
- Students will be tasked with applying the chosen method using actual leftover food brought from home (if applicable) or simulated leftovers provided by the teacher.

- They will follow step-by-step guidance on how to conserve the food according to their selected method. For example, if they chose refrigeration, they will organize and store the food correctly; if they picked freezing, they will seal and label the items.

Step 3: Report Writing (5 minutes)
- Students will write a short report documenting their chosen method and how they implemented it. This should include:
- The type of leftover food used.
- The method applied.
- Any challenges faced during the process.
- The outcome (how the food looked, smelled, etc.) after conservation.

Step 4: Presentation (5 minutes)
- Groups will take turns presenting their reports to the class. Encourage peer feedback and questions to foster a collaborative learning environment.

Conclusion (5 minutes)
- Summarize Key Points: Recap the methods discussed and the importance of conserving food.
- Interactive Activity: Conduct a quick quiz or game to reinforce the main ideas and ensure understanding.
- Preview Next Session: Briefly introduce the upcoming topics related to food sustainability or environmental impacts of food waste.

Extended Activities:
1. Home Assignment: Encourage students to implement their chosen food conservation method at home for a week and keep a journal of their experiences.
2. Research Project: Assign students to research a specific food preservation method (e.g., canning, dehydrating) and present their findings in the next class.
3. Community Involvement: Propose that students volunteer at a local food bank or soup kitchen to help manage food waste and understand community needs regarding food conservation.

Teacher Self-Evaluation:

WEEK 4: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Leftovers Food

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
1. Identify the methods of preparing leftover food for consumption to avoid wastage.
2. Discuss reheating as a method of preparing leftover food for consumption to avoid wastage.
3. Embrace reheating as a method of preparing leftover food for consumption.

Key Inquiry Question:
How is leftover food prepared for use to prevent food wastage?

Learning Resources:
- Lesson notes
- Digital devices
- Agriculture and Nutrition Learner's Textbook

Organization of Learning:

Introduction (5 minutes)
- Begin with a review of the previous lesson on food resources and their importance.
- Introduce the topic of leftover food.
- Guide learners to read and discuss relevant sections from the learning resources, focusing on the importance of conserving food.

Lesson Development (30 minutes)
In this section, students will work in pairs or small groups to explore methods of preparing leftover food:

Step 1: Brainstorming Methods (10 minutes)
- Have learners brainstorm various methods for preparing and using leftover foods (e.g., soups, casseroles, stir-fries).
- Each group will list these methods on a digital platform or write them on paper for sharing later.

Step 2: Research on Reheating (10 minutes)
- Instruct students to use digital or print resources to find information about reheating food.

- They should pay attention to safe reheating practices, such as temperature requirements and time guidelines.

Step 3: Group Discussion (5 minutes)
- Groups will discuss their findings and share insights on reheating methods.
- Encourage a short dialogue about why proper reheating is important for food safety and quality.

Step 4: Outline Benefits (5 minutes)
- Each group will outline the benefits of reheating food properly, such as preventing foodborne illnesses, enhancing flavors, and minimizing waste.
- Groups will present their key points briefly to the class for a collaborative learning experience.

Conclusion (5 minutes)
- Summarize the key findings and concepts discussed in the lesson.
- Conduct an interactive activity where students identify foods that can be successfully reheated, encouraging participation.
- Prepare students for the next session by posing questions to consider, such as “What other ways can we conserve food resources?”

Extended Activities:
1. Creative Cooking Challenge: Students can be challenged to create a new recipe using leftovers at home, document the process, and share the results with the class.

2. Food Waste Audit: Learners could track their food waste for a week and discuss their findings in the next lesson. This could lead to discussions about sustainable practices.

3. Reheating Techniques Presentation: Each pair could be tasked with creating a short presentation about a specific reheating method, demonstrating safe practices with visuals or videos.

Teacher Self-Evaluation:

WEEK 5: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Leftover Food

Specific Learning Outcomes:
- By the end of the lesson, students should be able to:
- Prepare leftover food for consumption using the reheating method to avoid wastage.
- Embrace reheating of leftover food for consumption at home to avoid wastage.

Key Inquiry Question(s):
- Why is proper reheating of leftover food important?

Learning Resources:
- Kitchen
- Leftover Food
- Digital devices

Organisation of Learning:

Introduction (5 minutes)
- Review Previous Lesson: Start the class by revisiting the last topic covered, discussing the importance of resource conservation in agriculture.
- Reading & Discussion: Provide learners with relevant material regarding food waste and reheating techniques. Encourage them to discuss and share their thoughts on key concepts.

Lesson Development (30 minutes)

Step 1: Introduction to Reheating Methods (10 minutes)
- Explain different reheating methods (microwave, stove, oven) and their benefits for safety and food quality.
- Discuss the importance of proper reheating for food safety, emphasizing reducing food waste and conserving resources.

Step 2: Group Preparation (10 minutes)
- Divide the class into small groups. Assign each group a type of leftover food that requires reheating.
- Instruct each group to choose a reheating method and begin preparing the leftover food. Encourage students to take notes or record the process on their devices.

Step 3: Reporting and Reflection (5 minutes)
- After preparation, guide students to write a short report on their experience, including any challenges faced and how they overcame them.
- Encourage reflection on how reheating the leftovers can help in reducing food waste.

Step 4: Presentations (5 minutes)
- Each group presents their reheated leftover dish to the class along with the reports they created. Encourage questions and discussions after each presentation.

Conclusion (5 minutes)
- Summarize Key Points: Recap the importance of reheating leftovers properly and how it contributes to conservation of resources.
- Interactive Activity: Conduct a quick quiz or a game whereby students identify best practices in reheating food.
- Preview Next Session: Introduce the upcoming topic on food preservation techniques, asking students to think about how they can store food safely.

Extended Activities:
- Cooking Challenge: Assign students to come up with a recipe that includes using leftover food and present it in the next class. They can research and create a simple booklet.
- Food Waste Audit: Encourage students to conduct a food waste audit at home for one week and report back on what types of food are wasted most and why.
- Community Service: Organize a food drive where students can collect non-perishable food items and prepare meals for a local food bank using leftover food approaches.

Teacher Self-Evaluation:

WEEK 5: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Leftovers Food

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Define the term recipe.
2. Describe preparing another recipe as a method of preparing leftover food for consumption to avoid wastage.
3. Embrace the preparation of another recipe as a method of preparing leftover food for consumption.

Key Inquiry Questions:
- What are the benefits of preparing another recipe as a method of preparing leftover food for consumption?

Learning Resources:
- Lesson notes
- Digital devices
- Agriculture and Nutrition Learner's Textbook

Organisation of Learning:

Introduction (5 minutes)
1. Begin by briefly reviewing the previous lesson on food waste.
2. Ask students what they believe a “recipe” is and record their responses.
3. Guide students to read and discuss the relevant content from their textbooks on recipes and food conservation, emphasizing the importance of not wasting food.

Lesson Development (30 minutes)

Step 1: Defining a Recipe (10 minutes)
- In pairs, learners will brainstorm what they think the word "recipe" means.
- Each pair will present their definition to the class.
- Teacher facilitates a discussion to consolidate understanding and provide a clear definition.

Step 2: Researching New Recipes (10 minutes)
- Students are grouped and asked to use digital or print resources (textbooks, internet) to find creative ways to use common leftover foods (like bread, rice, or vegetables).

- Each group will prepare short notes on at least one new recipe they can create using leftovers.

Step 3: Discussion on Benefits (5 minutes)
- In their groups, students discuss and list down the benefits of using leftovers in new recipes.
- Key points may include saving money, reducing food waste, creativity in cooking, and nutritional benefits.
- Groups share their lists with the class.

Step 4: Presentation and Wrap Up (5 minutes)
- Select a group to present their new recipe and the benefits discussed.
- Encourage the rest of the class to ask questions and provide feedback.

Conclusion (5 minutes)
- Summarize the importance of recipes and the practice of using leftovers to prevent food waste.
- Conduct a brief interactive activity, such as a group quiz or matching game with terms related to recipes and food conservation, to reinforce learning.
- Preview the next lesson, which will cover food preservation techniques.

Extended Activities:
- Have students create a "Leftover Recipe Book" where they compile their own family's practices or new recipes using leftovers.
- Encourage them to prepare a dish at home with parent involvement and bring in photos or reviews of their cooking experience to share in class.
- Organize a classroom challenge where students have to find the most creative way to use something they usually waste at home.

Teacher Self-Evaluation:

WEEK 5: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Leftover Food

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Outline recipes that can be adapted from leftover food to avoid food wastage.
2. Create their own recipes using common leftover food items.
3. Search online for recipes that utilize leftover food.
4. Embrace diverse recipes that help in minimizing food wastage.

Key Inquiry Question:
- Which recipes can you adopt from leftover food to avoid food wastage?

Learning Resources:
- Agriculture and Nutrition Learner's Textbook Recipes
- Digital devices (laptops, tablets, or smartphones)
- Lesson notes

Organisation of Learning:

Introduction (5 minutes)
- Briefly review the previous lesson content on food conservation.
- Ask students to share thoughts on why it’s important to reduce food waste.
- Guide them through relevant sections of the learning resources and highlight key concepts on using leftover food.

Lesson Development (30 minutes)
Step 1: Discussion
- In pairs, have learners discuss examples of common leftovers they encounter (e.g., cooked rice, vegetables, bread).
- Each pair shares their examples with the class.

Step 2: Recipe Listing
- As a class, create a list of possible recipes that can be made from the identified leftovers.
- Encourage creativity and consideration of local foods and resources.

Step 3: Online Research

- Using digital devices, learners will search the internet for additional recipes that utilize leftover food.
- Each group will find at least three new recipes and prepare to share them.

Step 4: Presentation
- Groups will present their findings, showcasing one or two of their favorite recipes and explaining how these recipes minimize food wastage.
- A gallery walk can be conducted where their recipes are displayed on posters around the classroom for peer review.

Conclusion (5 minutes)
- Summarize the key points discussed during the class, emphasizing the importance of using leftovers and reducing food waste.
- Conduct a brief interactive activity where students suggest modifications to a common leftover recipe.
- Preview the next lesson focusing on sustainable agricultural practices and how they tie into conserving food resources.

Extended Activities:
1. Recipe Book Creation: As a long-term project, students could compile a recipe book of leftover food recipes to encourage continuing this practice at home.
2. Food Waste Diary: Over a week, students can keep a diary of any food waste they produce and suggest ways to reduce it based on what they learned during the lesson.
3. Cooking Demonstration: Organize a class cooking demonstration where students can prepare one of the discussed recipes using leftovers, promoting hands-on learning.

Teacher Self-Evaluation:

WEEK 5: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Leftover Food

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Select a recipe that can be made from common leftover food at home.
2. Prepare leftover food for consumption by using another recipe method to avoid wastage.
3. Enjoy the process of preparing leftover food using the repurposing method.

Key Inquiry Question:
- How is leftover food prepared for use to prevent food wastage?

Learning Resources:
- Cooking tools and equipment
- Recipes
- Leftover food
- Digital devices (tablets or computers for research)

Organisation of Learning

Introduction (5 minutes)
- Review the previous lesson on food waste and its impact on the environment.
- Ask students to share what they remember about ways to use leftover food and guide them to read and discuss relevant content from handouts or digital resources, highlighting important concepts about repurposing food.

Lesson Development (30 minutes)

Step 1: Recipe Identification (10 minutes)
- In groups of 3-4, students will brainstorm common leftover foods they often have at home (e.g., vegetables, rice, bread).
- Each group will choose one type of leftover food and find a recipe that uses it. Encourage students to use digital devices for research and provide them with a list of simple recipes to consider.

Step 2: Collaborative Planning (10 minutes)
- Students will discuss their chosen recipe within their groups and write down the steps needed to prepare the leftover food.

- Groups will also plan how they can modify the recipe to enhance flavor or nutrition, encouraging creativity and critical thinking about ingredients.

Step 3: Preparation Activity (10 minutes)
- Using the cooking tools and leftover food provided, students will prepare their selected recipe.
- Emphasize teamwork, safety in the kitchen, and the importance of minimizing food waste during preparation.

Step 4: Taste Test and Feedback (5 minutes)
- After preparation, each group will present their dish to the class.
- Students will taste each other's creations and provide positive feedback. Discuss how these dishes could help reduce food waste at home.

Conclusion (5 minutes)
- Summarize key points: the significance of repurposing leftover foods, the creative process of finding new recipes, and the skills learned during the cooking activity.
- Conduct a fun interactive quiz or game where students can match leftover foods with potential recipes to reinforce what they've learned.
- Briefly preview the next session on food preservation techniques and ask students to think about foods they often waste at home.

Extended Activities:
1. Waste Audit Project: Have students conduct a week-long audit of the food waste in their homes and then create a plan to reduce it based on what they learned in class.
2. Family Recipe Collaboration: Encourage students to talk to family members about traditional recipes that use leftovers and create a class cookbook of these recipes.
3. Community Cooking Event: Organize a community potluck where students can bring dishes made from leftovers, promoting awareness about food waste in the community.

Teacher Self-Evaluation:

WEEK 6: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Conserving Leftover Food

Specific Learning Outcomes:
By the end of the lesson, learners will be able to:
1. State the importance of food safety when handling and storing leftover food.
2. Discuss the importance of food safety when handling and storing leftover food.
3. Acknowledge the benefits of food safety when handling and storing leftover food.

Key Inquiry Question:
- What is the importance of food safety when handling and storing leftover food?

Learning Resources:
- Lesson notes
- Digital devices (tablets/laptops)
- Agriculture and Nutrition Learner's Textbook
- Flashcards and posters related to food safety

Organisation of Learning:

Introduction (5 minutes)
1. Review the previous lesson's key points concerning food resources and conservation.
2. Introduce the topic of the day—food safety with leftovers—by reading aloud relevant sections from the textbook and encouraging students to discuss their previous experiences with leftovers.

Lesson Development (30 minutes)

Step 1: Brainstorming the Importance of Food Safety (10 minutes)
- In groups, students brainstorm key points about the importance of food safety. Each group will present their findings. Focus questions: Why is it crucial to handle leftover food safely? What risks are involved?

Step 2: Researching with Digital Resources (10 minutes)
- Students will use digital devices to research the importance of food safety. They should find statistics or articles related to foodborne illnesses caused by improper handling and storage of leftovers. Afterward, they will share interesting facts found with their peers.

Step 3: Group Discussion on Benefits (5 minutes)

- In their groups, learners discuss the benefits of practicing food safety with leftovers. They will explore issues like reducing waste, saving money, and promoting health.

Step 4: Presentation and Reflection (5 minutes)
- Each group will present their conclusions on the benefits and importance of food safety when handling and storing leftovers. Conclude this stage by highlighting recurring themes from presentations.

Conclusion (5 minutes)
- Summarize the key points discussed during the lesson regarding food safety and its importance.
- Conduct a brief interactive quiz or game (e.g., true/false or multiple-choice questions) to reinforce understanding of food safety practices.
- Prepare students for the next session by previewing the importance of meal planning and how to safely use leftovers.

Extended Activities:
1. Create a Food Safety Poster: Students create educational posters about food safety that can be displayed in the classroom or shared online.
2. Cooking Challenge: Plan a cooking session where learners must use leftovers creatively in a recipe, considering food safety guidelines.
3. Community Awareness Campaign: Students can develop a campaign to inform their families or community about food safety practices concerning leftovers.

Teacher Self-Evaluation:

WEEK 6: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Integrated Farming

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. State the meaning of integrated farming.
2. Discuss the importance of integrated farming in the conservation of resources.
3. Search the internet or textbook for information on the importance of integrated farming in conserving resources.
4. Appreciate the importance of integrated farming in conservation efforts.

Key Inquiry Questions:
1. What is Integrated Farming?
2. What is the importance of integrated farming in conserving resources?

Learning Resources:
- Lesson notes
- Digital devices (computers/tablets)
- Agriculture and Nutrition Learner's Textbook

Organisation of Learning

Introduction (5 minutes)
- Begin with a brief review of the previous lesson related to agricultural practices.
- Guide learners to read and discuss relevant content from the Agriculture and Nutrition Learner's Textbook, emphasizing the definition of integrated farming.

Lesson Development (30 minutes)

Step 1: Definition of Integrated Farming (10 minutes)
- In groups of 3-4, learners will use digital devices or print textbooks to find and write down the definition of integrated farming.
- Teacher circulates to provide support and answer questions.

Step 2: Importance of Integrated Farming (10 minutes)
- Learners will search for information (either in textbooks or online) about the benefits of integrated farming in conserving resources.

- Each group should come up with at least three key points related to the significance of integrated farming.

Step 3: Group Discussion (5 minutes)
- Groups will share their findings with the class, discussing how integrated farming helps in conserving resources like water, soil, and biodiversity.
- Teacher encourages students to ask questions and elaborate on each group's information.

Step 4: Reflection and Appreciation (5 minutes)
- As a class, reflect on the importance of integrated farming and how it directly contributes to sustainable agriculture.
- Ask students to think about their personal views on integrated farming and share any additional thoughts.

Conclusion (5 minutes)
- Summarize the key points discussed during the lesson: definition of integrated farming and its importance in conservation.
- Conduct a quick interactive quiz (e.g., Kahoot!) to reinforce key topics.
- Provide a preview of upcoming lessons, highlighting the next topic of discussion.

Extended Activities
- Research Project: Assign learners a mini-project where they research a local farm and analyze how integrated farming methods might be applied there, focusing on resource conservation.
- Field Trip: Organize a visit to a local farm that practices integrated farming for hands-on learning.
- Presentation: Encourage groups to create a presentation on integrated farming systems featuring case studies of successful practices in different areas.

Teacher Self-Evaluation:

WEEK 6: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Integrated Farming

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- Identify the components of integrated farming in conserving resources.
- Describe crop production as a component of integrated farming and its contribution to resource conservation.
- Acknowledge the importance of crop production as a component of integrated farming in resource conservation.

Key Inquiry Question(s):
- What are the components of Integrated Farming in conserving resources?

Learning Resources:
- Agriculture and Nutrition Learner's Textbook
- Lesson notes
- Digital devices (tablets, laptops)
- Resource person (guest speaker or local farmer)

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on resource conservation.
- Guide learners in reading and discussing relevant content from the learning resources, focusing on understanding key concepts of integrated farming.

Lesson Development (30 minutes)

Step 1: Introduction to Integrated Farming (10 minutes)
- Briefly introduce integrated farming and its purpose in resource conservation.
- Discuss various components of integrated farming, such as crop production, livestock, and aquaculture.

Step 2: Group Research on Components (10 minutes)
- In groups or pairs, learners use digital or print resources to research the different components of integrated farming.

- Each group should aim to identify at least three components and their roles in resource conservation.

Step 3: Focus on Crop Production (5 minutes)
- Reconvene as a class, and each group shares their findings on the components of integrated farming.
- Lead a discussion that specifically details crop production, highlighting its practices and how it contributes to conserving resources (e.g., soil health, water usage).

Step 4: Importance of Crop Production (5 minutes)
- Discuss as a class why crop production is crucial in an integrated farming system.
- Ask students to brainstorm ways in which improving crop production processes can lead to better resource conservation.

Conclusion (5 minutes)
- Summarize the key points discussed and the specific learning objectives achieved during the lesson.
- Conduct a brief interactive activity, such as a quick quiz or a group discussion on how integrated farming could be applied in their local area.
- Preview upcoming topics: The students will explore the interaction between livestock and crop production in integrated systems.

Extended Activities:
- Assign students to create a poster or presentation on a specific component of integrated farming (e.g., aquaculture or livestock management) and its impact on resource conservation.
- Encourage students to visit a local farm or community garden, then report on the use of integrated farming practices observed.
- Have students write a reflection on what practices they can adopt at home to conserve resources based on integrated farming principles.

Teacher Self-Evaluation:

WEEK 6: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Integrated Farming

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
1. Outline the contribution of livestock farming in conservation of resources as a component of integrated farming.
2. Describe livestock farming as a component of integrated farming.
3. Appreciate livestock farming as a component of integrated farming in conserving resources.

Key Inquiry Question(s):
- How does livestock farming help in conserving resources in an integrated farm?

Learning Resources:
- Lesson notes
- Digital resources
- Resource person (Agriculture and Nutrition Learner's Textbook)

Organisation of Learning

Introduction (5 minutes):
- Review the previous lesson on integrated farming and its general principles.
- Ask students to share any examples of integrated farming they might know, focusing on livestock and crop interaction.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing key concepts related to livestock farming.

Lesson Development (30 minutes):

Step 1: Identifying Livestock and Crops (10 minutes)
- In groups or pairs, students will brainstorm and list examples of livestock and crops that can be raised together on an integrated farm.
- Each group will present their list to the class for discussion.

Step 2: Research on Livestock Farming (10 minutes)
- Using digital or print resources, students will conduct research on the following aspects:
- Types of livestock suitable for integrated farming.
- Benefits of having livestock alongside crops.

- Each group will prepare a few sentences to share their findings with the class.

Step 3: Discussion on Resource Conservation (5 minutes)
- Students will discuss how livestock farming contributes to conserving resources such as soil, water, and nutrients in an integrated farming system.
- Teachers will guide the discussion with prompting questions like: “How do animals benefit crop production?” and “What resources can be saved with livestock in an integrated system?”

Step 4: Presenting Findings (5 minutes)
- Groups will present their discussions regarding the contribution of livestock to resource conservation.
- Encourage questions and interaction from other students to deepen understanding of how livestock integrates into conservation practices.

Conclusion (5 minutes):
- Summarize the key points discussed during the lesson, focusing on the relationship between livestock farming and resource conservation in integrated systems.
- Conduct a brief interactive activity such as a Quiz or a Poll related to the key concepts discussed.
- Prepare learners for the next session with a preview of upcoming topics or questions to consider regarding other components of integrated farming.

Extended Activities:
1. Research Project: Have students select a specific type of livestock and research its role in integrated farming, presenting their findings to the class in a short presentation.
2. Field Trip: Organize a visit to a local integrated farm, if possible, where students can observe livestock and crops coexisting and conserving resources.
3. Reflection Journal: Ask students to maintain a journal where they describe how they would implement an integrated farm on their own, considering livestock and crop choices.

Teacher Self-Evaluation:

WEEK 7: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Integrated Farming

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
1. Outline the contribution of fish farming as a component of integrated farming in conserving resources.
2. Describe fish farming (aquaculture) as a component of integrated farming in conserving resources.
3. Acknowledge the benefits of aquaculture in the conservation of resources.

Key Inquiry Question(s):
- How does fish farming help in conserving resources?

Learning Resources:
- Lesson notes
- Digital resources
- Resource person (guest speaker)
- Agriculture and Nutrition Learner's Textbook

Organisation of Learning

Introduction (5 minutes)
- Review the previous lesson focused on the principles of integrated farming.
- Engage learners in a discussion by asking what they remember about integrating different farming practices and the benefits they can have on conservation.

Lesson Development (30 minutes)
- Divide the lesson into the following distinct steps:

Step 1: Introduction to Fish Farming
- Introduce the concept of fish farming (aquaculture). Explain its definition and its role in agriculture as part of integrated farming.
- Present data or facts about global fish production and its importance in food security.

Step 2: Researching Resources

- In pairs or small groups, learners use digital or print resources to investigate how fish farming is implemented and its role in conserving resources such as water, land, and nutrients.
- Provide guiding questions to help them focus their research such as: What methods are used in fish farming? How does it reduce reliance on land-based feed and increase efficient resource use?

Step 3: Discussion of Findings
- Once research is completed, each group discusses their findings and prepares to share how fish farming contributes to resource conservation. Encourage them to think critically about the sustainability aspect.
- Provide a framework for presenting their findings, including key points to cover such as ecological benefits, economic contributions, and effects on local communities.

Step 4: Class Presentation
- Groups present their findings to the class. Encourage interaction and questions from the audience to foster a collaborative learning environment.
- As groups present, highlight key contributions to resource conservation and compile a list on the board.

Conclusion (5 minutes)
- Summarize the key points discussed during the lesson, emphasizing the role of fish farming in conserving resources and the interconnectedness of farming practices.
- Conduct a brief interactive activity, such as a quick quiz or a think-pair-share where students collaborate on future applications of aquaculture in conservation.
- Prepare learners for the next session by previewing related topics, such as sustainable farming practices or the role of aquaculture in climate resilience.

Extended Activities:
- Encourage learners to create a small project where they design a model of an integrated farm that includes aquaculture and present its benefits and resource conservation strategies.
- Organize a field trip to a local fish farm or aquaculture facility to observe real-world applications and conservation methods.
- Assign a research paper on the impact of fish farming on local ecosystems, requiring them to interview a local aquaculture expert or farmer.

Teacher Self-Evaluation:

WEEK 7: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Integrated Farming

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- State the contribution of agroforestry in conserving resources in the environment.
- Describe agroforestry as a component of integrated farming in conserving resources.
- Acknowledge the importance of agroforestry in the conservation of resources in the environment.

Key Inquiry Questions:
- What is Agroforestry?
- How does Agroforestry contribute to conserving resources in the environment?

Learning Resources:
- Lesson notes.
- Digital resources.
- Agriculture and Nutrition Learner's Textbook.

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on integrated farming systems.
- Guide learners to read and discuss content from their learning resources, focusing on key concepts related to agroforestry.

Lesson Development (30 minutes)

Step 1: Define Agroforestry
- In groups, learners will discuss and create a definition of agroforestry. They should consider what it means and how it relates to farming practices. Groups will then share their definitions with the class.

Step 2: Identify Examples
- Learners will list examples of trees used in agroforestry systems. They may choose to identify both native and introduced species of trees that are beneficial in agricultural settings.

Step 3: Research the Contribution of Agroforestry

- Using digital or print resources, groups will research how agroforestry contributes to resource conservation. They should focus on aspects such as soil conservation, biodiversity enhancement, and climate resilience.

Step 4: Class Discussion and Presentation
- Each group will discuss their findings about the contributions of agroforestry in conserving resources. They will present their key points to the class, encouraging questions and discussions.

Conclusion (5 minutes)
- Summarize the key points discussed during the lesson, affirming the learning objectives achieved.
- Conduct a brief interactive quiz or game to review the main concepts covered.
- Preview the next session, which will delve into specific agroforestry practices and their impact on sustainability.

Extended Activities:
- Field Research Project: Encourage students to visit a local agroforestry site or community garden, observe the practices, and report back on the benefits of the trees used in terms of resource conservation.
- Create an Agroforestry Plan: Have students design a simple agroforestry plan for a small plot of land, detailing tree types, crops to be grown, and how they will promote conservation.
- Debate: Organize a debate on the advantages and challenges of implementing agroforestry in modern farming practices.
- Guest Speaker: Arrange for a local farmer or agricultural expert to speak about their experiences with agroforestry.

Teacher Self-Evaluation:

WEEK 7: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Integrated Farming

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. State the contribution of rabbit and poultry keeping as components of integrated farming in conserving resources.
2. Describe rabbit and poultry keeping as components of integrated farming.
3. Embrace poultry and rabbit keeping as components of integrated farming in conserving resources.

Key Inquiry Question:
- What are the contributions of rabbit and poultry keeping in conserving resources?

Learning Resources:
- Lesson notes
- Digital resources
- Agriculture and Nutrition Learner's Textbook
- Resource person

Organisation of Learning

Introduction (5 minutes)
- Begin with a brief review of the previous lesson to refresh students' memories.
- Guide learners to read relevant sections from the learning resources, prompting them to discuss key concepts.

Lesson Development (30 minutes)

Step 1: Introduction to Integrated Farming (10 minutes)
- Introduce the concept of integrated farming and its benefits.
- Explain how rabbit and poultry keeping fits into this system. Discuss their roles in resource conservation.

Step 2: Group Discussion on Rabbit and Poultry Keeping (10 minutes)
- Divide the class into small groups or pairs.
- Task each group with discussing the characteristics and benefits of rabbit and poultry keeping. Encourage sharing of ideas on how these animals contribute to resource conservation.

Step 3: Research Activity (5 minutes)
- Allow groups to use digital or print resources to find information on specific benefits of poultry and rabbit keeping in resource conservation (e.g., manure production, pest control).

Step 4: Presentation of Findings (5 minutes)
- Invite each group to present their findings on the contributions of rabbit and poultry keeping in conserving resources.
- Facilitate a brief discussion on the points raised, encouraging other students to ask questions or provide additional insights.

Conclusion (5 minutes)
- Summarize the key points discussed during the lesson, reinforcing the learning objectives achieved.
- Conduct a brief interactive quiz or activity (e.g., a true/false game or a quick response round) to reinforce the main topics.
- Prepare learners for the next session by previewing upcoming topics or posing questions for them to think about related to integrated farming.

Extended Activities
- Field Trip: Plan a visit to a local farm practicing integrated farming. Students can observe rabbit and poultry keeping in action and see first-hand their contributions to resource conservation.
- Project Assignments: Assign students to create a presentation or poster on different types of integrated farming practices worldwide, focusing on the specifics of rabbit and poultry keeping.
- Research Paper: Encourage students to write a short paper on the environmental benefits of integrated farming and suggest ways to improve resource conservation on local farms.

Teacher Self-Evaluation:

WEEK 7: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Integrated Farming

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Identify the vegetables that can be grown in an integrated farm.
2. Discuss the contribution of vegetable production in conserving resources in an integrated farm.
3. Embrace vegetable production as a component of integrated farming in conserving resources.

Key Inquiry Question(s):
- How does vegetable production help in conserving resources in an integrated farm?

Learning Resources:
- Lesson notes
- Agriculture and Nutrition Learner's Textbook
- Digital resources (online articles, videos)
- Resource person (guest speaker from local farms)

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson briefly, engaging students to recall key points.
- Guide learners to skim through the relevant chapters in the Agriculture and Nutrition textbook and discuss the importance of vegetable production in integrated farming.

Lesson Development (30 minutes)

Step 1: Listing Vegetables (10 minutes)
- In pairs, learners brainstorm and list at least five types of vegetables that can be grown in an integrated farm.
- Invite each pair to share their lists with the class, and compile a comprehensive list on the board.

Step 2: Research and Discussion (10 minutes)
- Assign groups each a specific vegetable from the compiled list to research.
- Using digital or print resources, each group will find information on:
- The growing conditions for their assigned vegetable

- How growing this vegetable might help conserve resources (e.g., water, soil health) in an integrated farm.
- Groups prepare for a short class presentation.

Step 3: Presentations (5 minutes)
- Each group presents their findings, explaining how their chosen vegetable contributes to resource conservation within integrated farming practices.
- Encourage questions and discussion from peers after each presentation.

Step 4: Reflection (5 minutes)
- Engage the class in a reflective conversation:
- How do the presented vegetables contribute to the sustainability of integrated farming?
- What new understandings do students have about the role of vegetable production?

Conclusion (5 minutes)
- Summarize key points and insights about vegetable production and its role in conserving resources.
- Conduct a brief interactive activity (e.g., a quiz or brainstorming session) to reinforce the main topics discussed.
- Prepare learners for the next session by previewing topics related to crop rotation or nutrient management.

Extended Activities:
- Research Project: Students can create a poster or a digital presentation on a specific integrated farming system that uses various vegetables and their roles in resource conservation.
- Guest Speaker: Invite a local farmer who practices integrated farming to discuss their experiences and challenges, enhancing the connection between theory and practice.
- Field Trip: Organize a visit to a local integrated farm to observe firsthand the practical application of growing vegetables within an integrated system.

Teacher Self-Evaluation:

WEEK 9: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Integrated Farming

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Outline the relational benefits of vegetable production, poultry, and rabbit keeping in conserving resources.
- Discuss the relational benefits of vegetable production, rabbit, and poultry keeping in conserving resources.
- Acknowledge the relational benefits of vegetable production, rabbit, and poultry keeping in an integrated farm.

Key Inquiry Question(s):
- How does vegetable production, poultry, and rabbit keeping work together in conserving resources in an integrated farm?

Learning Resources:
- Lesson notes
- Digital devices
- Agriculture and Nutrition Learner's Textbook

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on agriculture sustainability and resource management.
- Guide learners to read and discuss relevant content from the learning resources, ensuring they comprehend the significance of integrated farming systems.

Lesson Development (30 minutes)

Step 1: Identifying Benefits
- In groups of 4, learners will list the benefits of vegetable production, poultry, and rabbit keeping.
- Each group will focus on one farming practice and how it contributes to conserving resources, such as reducing waste and maximizing yields.

Step 2: Research Activity

- Using digital devices or textbooks, groups will research additional benefits and examples of integrated farming systems that successfully combine these three practices.
- Encourage students to look for real-life examples or case studies to support their findings.

Step 3: Group Discussion
- Groups will come together to discuss their findings. Each group will present the interconnected benefits of their assigned farming practice and how it relates to the other two.
- Promote critical thinking by facilitating questions around resource conservation and sustainability.

Step 4: Creative Integration
- As a collaborative activity, groups will create a visual poster or diagram that illustrates the interconnected relationships between vegetable production, poultry, and rabbit keeping in an integrated farm.
- Groups will present their visuals to the class, highlighting key points of resource conservation.

Conclusion (5 minutes)
- Summarize the key points discussed in the lesson about the relational benefits of integrated farming.
- Conduct a brief interactive activity, such as a quick quiz or group reflection, to reinforce the main topics.
- Preview upcoming topics, such as sustainable practices in farming or advanced integrated farming techniques, and pose questions to consider at home, encouraging them to think of examples in their local community.

Extended Activities:
1. Farm Tour Project: Plan a visit to a local farm that utilizes integrated farming practices or research one online. Create a report detailing how resources are conserved on-site.
2. Infographic Creation: Use digital tools to create an infographic summarizing the benefits of integrated farming and resource conservation for a school presentation.
3. Home Experiment: Encourage students to start a small integrated garden project at home, combining vegetables with herbs and small animals if possible, and keep a diary of their observations on resource conservation.

Teacher Self-Evaluation:

WEEK 9: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Integrated Farming

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Outline the benefits of organic waste management and water management systems as components of integrated farming in conserving resources.
2. Discuss the contribution of organic waste management and water management systems in conserving resources in an integrated farm.
3. Embrace organic waste management and water management systems in conserving resources in an integrated farm.

Key Inquiry Question:
How does organic waste management and water management system contribute to the conservation of resources in an integrated farm?

Learning Resources:
- Lesson notes
- Digital resources
- Agriculture and Nutrition Learner's Textbook
- Resource person

Organisation of Learning:

Introduction (5 minutes)
- Begin with a review of the previous lesson on sustainable farming practices.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing the understanding of key concepts related to organic waste and water management.

Lesson Development (30 minutes)

Step 1: Introduction to Organic Waste Management (10 minutes)
- Explain the concept of organic waste management.
- Discuss types of organic wastes (e.g., crop residues, animal manure) and how they can be recycled on the farm.
- Ask learners to share their thoughts on the importance of recycling organic waste.

Step 2: Introduction to Water Management Systems (10 minutes)

- Describe what water management systems are and their importance in agriculture.
- Discuss techniques such as rainwater harvesting, drip irrigation, and water recycling.
- Encourage learners to think about how these systems can help in resource conservation.

Step 3: Group Activity (5 minutes)
- Divide learners into small groups or pairs.
- Task each group to brainstorm ways in which organic waste management and water management systems can be integrated on a farm to conserve resources.
- Have groups prepare to share their ideas with the class.

Step 4: Class Discussion (5 minutes)
- Invite groups to present their findings and discuss how integrating these systems helps to conserve resources.
- Facilitate a class discussion about the benefits and any challenges they might encounter.

Conclusion (5 minutes)
- Summarize key points discussed during the lesson, highlighting the advantages of organic waste management and water management systems in integrated farming.
- Conduct an interactive quiz or a brief discussion to reinforce learning.
- Prepare learners for the next session by introducing the upcoming topic of soil management.

Extended Activities:
1. Research Project: Have students research a local farm that uses integrated farming practices. Ask them to report on the methods of organic waste and water management used.
2. Field Trip: Plan a visit to a local organic farm so students can observe organic waste and water management systems in action.
3. Create a Resource Management Plan: Students can create a plan for managing organic waste and water resources for a hypothetical farm.

Teacher Self-Evaluation:

WEEK 9: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Integrated Farming

Specific Learning Outcomes
By the end of the lesson, the learner should be able to:
- Create a layout of their own integrated farm design on Manila paper to illustrate integrated farming components.
- Enjoy creating a layout of their integrated farm design.

Key Inquiry Question
- Where can you draw a layout of the integrated farm design?

Learning Resources
- Manila papers
- Pencils and erasers
- Drawing papers

Organisation of Learning

Introduction (5 minutes)
- Begin with a brief review of the previous lesson about the importance of integrated farming and its components.
- Guide learners to read and discuss content related to integrated farming, highlighting key concepts such as crop rotation, livestock integration, and sustainable practices.

Lesson Development (30 minutes)

Step 1: Understanding Integrated Farming (10 minutes)
- Discuss the main components of integrated farming (e.g., crops, livestock, aquaculture, etc.).
- Ask students to brainstorm what features might be included in an integrated farm design to optimize resource use.

Step 2: Group Brainstorming (10 minutes)
- Divide the class into small groups and ask them to discuss their ideas about what their integrated farm would look like.
- Each group should note down elements they want to incorporate into their design, focusing on balance and sustainability.

Step 3: Individual Layout Creation (10 minutes)
- Provide each student with Manila paper, pencils, and erasers.
- Guide them to draw their own integrated farm layout, including all the elements discussed, such as where crops will be planted, where livestock will be kept, etc.
- Encourage creativity and remind students to label the components of their designs.

Step 4: Peer Sharing and Feedback (5 minutes)
- Allow students to pair up and share their farm designs with one another.
- Encourage them to give constructive feedback on each other's designs focusing on sustainability and resource conservation.

Conclusion (5 minutes)
- Summarize the key points learned about integrated farming layouts and their components.
- Conduct an interactive activity: Ask students a few quick questions related to the components of their designs or why integrated farming is essential.
- Preview the upcoming lesson on resource conservation practices in integrated farming and pose questions for consideration like "What are the benefits of using compost in farming?"

Extended Activities
- Research Project: Assign learners to research a successful integrated farm model (local or international) and present its benefits and challenges.
- Field Trip: Plan a visit to a local farm practicing integrated farming to observe and ask questions.
- Journal Entry: Encourage students to keep a journal where they reflect on their learning about integrated farming and ideas they would like to implement in their own lives.

Teacher Self-Evaluation:

WEEK 9: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Conservation of Resources
Sub Strand: Integrated Farming

Specific Learning Outcomes
By the end of the lesson, learners should be able to:
1. Make a model to illustrate integrated farming components using locally available materials.
2. Value self and each other's effort in constructing a model of an integrated farm.

Key Inquiry Question
- Which locally available materials can you use to make a model of an integrated farm?

Learning Resources
- Cardboards
- Papers
- Manilla papers
- Sticking glues
- Cartons

Organisation of Learning

Introduction (5 minutes)
1. Review Previous Lesson: Begin the class by briefly discussing what was learned in the previous lesson about farming practices and their importance in resource conservation.
2. Discussion: Introduce the key concepts of integrated farming, discussing its benefits and how it utilizes various resources effectively. Encourage learners to participate by sharing their thoughts on what they know about integrated farming.

Lesson Development (30 minutes)

Step 1: Identify Materials (7 minutes)
- Guide learners to brainstorm and list locally available materials they could use to create their model. Ask students to think about everyday items they can find around their homes or school.

Step 2: Collect Materials (8 minutes)
- Allow learners to gather the suggested materials from their desks or classroom supplies to use for model building. Encourage collaboration by allowing them to team up to share resources.

Step 3: Model Construction (10 minutes)

- In pairs or small groups, learners will start building their models of an integrated farm. Provide guidance on how to represent different components (e.g., crops, livestock, water systems, etc.) and emphasize creativity.

Step 4: Class Presentations (5 minutes)
- After the models are constructed, give each pair or group a chance to present their model to the class. Encourage peer feedback by asking classmates to identify creative aspects and the effectiveness of resource use.

Conclusion (5 minutes)
1. Summarize Key Points: Recap the importance of integrated farming and how the models illustrated the various components involved.
2. Interactive Activity: Conduct a quick quiz or ask a few questions that allow students to reflect on what they've learned. For example, "What is one key benefit of integrated farming?"
3. Preview Next Session: Briefly discuss what will be learned in the next class, such as the importance of sustainable practices in agriculture, and pose questions for students to consider, such as, "How can we apply the concept of integrated farming in our communities?"

Extended Activities
1. Research Assignment: Have students research different types of integrated farming systems used in their region or globally and prepare a short report or presentation for the next class.
2. Create a Journal: Ask learners to keep a journal of local farms or small-scale producers they encounter over the next week. They should note down the farming practices they observe and evaluate their sustainability.

Teacher Self-Evaluation:

WEEK 10: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Food Production Processes
Sub Strand: Organic Gardening

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
- Define the term organic gardening in crop production.
- Discuss the importance of organic gardening in the production of healthy foods.
- Search for information on the importance of organic gardening in the production of healthy foods.
- Acknowledge the benefits of organic gardening in the production of healthy foods.

Key Inquiry Questions:
- What is organic gardening?
- Why should we practice organic gardening?

Learning Resources:
- Lesson notes
- Digital resources
- Resource person (Guest speaker, if applicable)
- Agriculture and Nutrition Learner's Textbook

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on crop production methods.
- Guide learners to read and discuss relevant content from the learning resources, focusing on organic practices in gardening and their key concepts.

Lesson Development (30 minutes)

Step 1: Definition of Organic Gardening (10 minutes)
- In pairs, learners will brainstorm and write down what they think organic gardening means.
- Discuss and refine their definitions, then share with the class.
- Teacher provides a clear definition and highlights key characteristics of organic gardening.

Step 2: Importance of Organic Gardening (10 minutes)
- Learners will work in small groups to search through digital or print resources for information on the importance of organic gardening in healthy food production.

- Each group will prepare a short presentation summarizing their findings, focusing on health benefits and environmental impact.

Step 3: Group Presentations (5 minutes)
- Groups will present their findings to the class.
- Encourage questions and discussions to deepen understanding.

Step 4: Acknowledging Benefits (5 minutes)
- As a class, compile a list of the benefits of organic gardening based on the presentations and discussions.
- Highlight key benefits identified, such as improved soil health, environmental sustainability, and better nutrition.

Conclusion (5 minutes)
- Summarize the key points discussed during the lesson, revisiting the definitions and benefits of organic gardening.
- Conduct a brief interactive activity: for example, a quick quiz or a 'think-pair-share' where students share one benefit of organic gardening with a partner.
- Prepare learners for the next session by previewing upcoming topics, such as composting or pest management in organic gardening.

Extended Activities:
- Encourage students to start their own small organic garden at home or in school. Have them keep a journal documenting their observations and learning process.
- Organize a field trip to a local organic farm to observe practices firsthand and engage with organic farmers.
- Create a presentation or digital poster about organic gardening methods, highlighting what they learned, and share it on social media or in a school exhibition.

Teacher Self-Evaluation:

WEEK 10: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Food Production Processes
Sub Strand: Organic Gardening

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Identify organic gardening practices in crop production.
- Discuss the use of organic manure as an organic gardening practice.
- Prepare organic manure from plant matter and animal waste.
- Acknowledge the benefits of organic manure in the production of healthy foods.

Key Inquiry Question(s):
- What organic gardening practices do you know?
- What are the benefits of organic manure in crop production?

Learning Resources:
- Lesson notes
- Digital resources
- Resource person
- Video clips
- Agriculture and Nutrition Learner's Textbook

Organisation of Learning:

Introduction (5 minutes):
1. Review the previous lesson by asking learners about what they learned. Capture their responses to gauge understanding.
2. Guide learners to read a short excerpt from the Agriculture and Nutrition Learner's Textbook about organic gardening, focusing on identifying practices and the role of organic manure.

Lesson Development (30 minutes):

Step 1: Identifying Organic Gardening Practices (10 minutes)
- In pairs, learners brainstorm and list as many organic gardening practices as they can think of. After 5 minutes, ask each pair to share one practice with the class. Compile a master list on the board.

Step 2: Discussing Organic Manure (10 minutes)

- Using digital resources, learners research the use of organic manure in crop production. They will discuss as a group:
* What is organic manure?
* Why is it different from chemical fertilizers?
* What are its benefits?
- Encourage learners to take notes on key points for class discussion.

Step 3: Preparing Organic Manure (5 minutes)
- Provide a simple visual guide on how to prepare organic manure.
- In groups, learners outline the steps needed to create organic manure from plant matter and animal waste.

Step 4: Hands-On Preparation (Optional depending on resources) (5 minutes)
- If time and resources allow, learners gather existing plant materials and animal waste (real or simulated) to demonstrate preparing organic manure.

Conclusion (5 minutes):
1. Summarize key points from the lesson, emphasizing organic practices and the preparation of organic manure.
2. Conduct a brief interactive activity, such as a quick quiz or a "think-pair-share" where students discuss one benefit of organic manure.
3. Prepare learners for the next session by hinting at themes like sustainable agriculture practices or pest management.

Extended Activities:
- Gardening Project: Initiate a class project where students create their own small organic gardens. They can track the growth of their plants and apply the organic manure they prepared.
- Research Assignment: Have students research a particular crop and explore how organic practices can enhance its production. Present findings in a future class.
- Field Trip: Plan a visit to a local organic farm or a community garden to observe organic gardening practices in action.

Teacher Self-Evaluation:

WEEK 10: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Food Production Processes
Sub Strand: Organic Gardening

Specific Learning Outcomes:
By the end of the lesson, the learner should be able to:
1. Explain the use of organic pesticides as an organic gardening practice in crop production.
2. Search for information on how to prepare an organic pesticide using locally available materials.
3. Use locally available materials to prepare an organic pesticide.
4. Acknowledge the benefits of organic pesticides in crop production.

Key Inquiry Question(s):
- What are the benefits of organic pesticides in crop production?

Learning Resources:
- Lesson notes.
- Digital resources (internet access for research).
- Agriculture and Nutrition learner's textbook.
- Resource person (guest speaker knowledgeable about organic farming).

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson on sustainable farming practices.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing the understanding of organic gardening practices and the role of organic pesticides.

Lesson Development (30 minutes)
Step 1 (10 minutes): Introduction to Organic Pesticides
- Teacher explains what organic pesticides are and how they differ from synthetic pesticides.
- Discuss the importance of organic pesticides in promoting sustainable agriculture and protecting the environment.

Step 2 (5 minutes): Research and Group Discussion
- In pairs or small groups, learners use digital or print resources to search for information on organic pesticides and their benefits in crop production.
- Each group discusses their findings, focusing on the effectiveness, safety, and environmental impact of organic pesticides.

Step 3 (10 minutes): Identifying Natural Ingredients
- Groups research online to identify natural substances available in their local environment that can be used to create organic pesticides.
- Share these findings with the class, recording them on a board or digital document.

Step 4 (5 minutes): Preparation of an Organic Pesticide
- Guided by the teacher, learners collaboratively select one natural substance from their research.
- Discuss how to prepare an organic pesticide using this ingredient. If possible, involve a resource person to demonstrate preparation on a small scale, or provide a simple recipe for learners to try at home.

Conclusion (5 minutes)
- Summarize key points regarding organic pesticides: their definition, examples of organic materials, benefits, and preparation methods.
- Conduct a brief interactive quiz or discussion to reinforce learning.
- Preview upcoming topics related to sustainable agriculture practices that learners should consider before the next class.

Extended Activities:
1. Personal Project: Ask learners to create a small home garden where they can apply organic pesticide practices and report their findings in the next class.

2. Research Assignment: Have students write a short essay on the benefits and limitations of organic versus synthetic pesticides in crop production, including interviews with local farmers.

3. Field Trip: Organize a visit to a local organic farm or garden to observe real-world applications of organic gardening practices, including pesticide use.

Teacher Self-Evaluation:

WEEK 10: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Food Production Processes
Sub Strand: Organic Gardening

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
1. Identify the mechanical methods of weed control.
2. Discuss mechanical weed control and the use of organic foliar feed made from animal wastes and plants as organic gardening practices.
3. Search the internet for the benefits of mechanical weed control and the use of organic foliar feed in crop production.
4. Embrace mechanical weed control and the use of organic foliar feed in crop production.

Key Inquiry Question(s):
- What are the benefits of mechanical weed control?
- What are the benefits of organic foliar feed in crop production?

Learning Resources:
- Agriculture and Nutrition Learner's Textbook
- Lesson notes
- Digital resources (internet access)
- Animal wastes and plants (for foliar feed preparation)

Organisation of Learning

Introduction (5 minutes)
- Review the previous lesson, focusing on key concepts related to organic gardening practices.
- Guide learners to read and discuss relevant content from the learning resources, emphasizing the understanding of mechanical weed control and organic foliar feeds.

Lesson Development (30 minutes)
Step 1: Identify Mechanical Weed Control Methods (10 minutes)
- In pairs, learners list common mechanical methods for weed control (e.g., hoeing, mulching, hand pulling).
- Share findings with the class and discuss the importance of these methods in organic gardening.

Step 2: Research Benefits (10 minutes)
- Using digital resources, learners search for the benefits of mechanical weed control and organic foliar feed.

- Each group selects one benefit to present to the class.

Step 3: Class Discussion (5 minutes)
- Open the floor for discussion where learners share their findings on the benefits of mechanical weed control and organic foliar feed.
- Encourage learners to ask questions and offer additional information.

Step 4: Prepare Organic Foliar Feed (5 minutes)
- Demonstrate how to prepare an organic foliar feed using animal wastes and plants (e.g., compost tea).
- If feasible, guide learners through the process, ensuring everyone understands how to create it for future use.

Conclusion (5 minutes)
- Summarize key points: the importance of mechanical weed control, the benefits of organic gardening practices, and the steps to prepare organic foliar feed.
- Conduct a brief interactive activity where learners quiz each other on one benefit of mechanical weed control and one method of preparing foliar feed.
- Preview the next session on more advanced organic gardening techniques, encouraging learners to think about the impact of their methods on sustainability.

Extended Activities:
1. Gardening Project: Have students start a small garden plot or container garden where they can apply mechanical weed control and create organic foliar feeds.
2. Research Assignment: Learners can research and present on various organic gardening practices used around the world.
3. Field Trip: Organize a visit to a local organic farm where students can see mechanical weed control methods and organic fertilization techniques in action.

Teacher Self-Evaluation:

WEEK 11: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Food Production Processes
Sub Strand: Organic Gardening

Specific Learning Outcomes:
- By the end of the lesson, the learner should be able to:
1. Outline the maintenance practices of crops grown using organic gardening.
2. Discuss the maintenance practices of crops grown using organic gardening.
3. Appreciate the maintenance practices of crops grown using organic gardening.

Key Inquiry Question(s):
- What are the maintenance practices of crops grown using organic gardening?

Learning Resources:
- Lesson notes
- Agriculture and Nutrition Learner's Textbook
- Posters
- Digital resources (videos, articles)

Organisation of Learning:

Introduction (5 minutes)
- Review the previous lesson's key concepts related to organic gardening.
- Guide learners in a brief reading session from the Agriculture and Nutrition Learner's Textbook and related posters, emphasizing essential maintenance practices of organic gardens.

Lesson Development (30 minutes)

Step 1: Brainstorming Session (10 minutes)
- Activity: Divide learners into small groups. Each group brainstorms and lists the maintenance practices they know related to organic gardening.
- Outcome: Each group shares one idea with the class, fostering open discussion.

Step 2: Research Time (10 minutes)
- Activity: Groups use digital or print resources to research specific maintenance practices of organic gardening (e.g., pest management, watering techniques, soil health).
- Outcome: Each group compiles their findings to prepare for sharing with the class.

Step 3: Group Discussion (5 minutes)

- Activity: Groups discuss their research findings, focusing on the importance of each maintenance practice.
- Outcome: Encourage students to connect why these practices are vital for the health of crops and the environment.

Step 4: Presentations (5 minutes)
- Activity: Each group presents their researched findings on maintenance practices to the class.
- Outcome: Students are given time to ask questions or provide feedback on the presentations.

Conclusion (5 minutes)
- Summarize key points regarding the maintenance practices of organic crops discussed in class.
- Conduct a brief interactive quiz or Kahoot to reinforce the main topics covered during the lesson.
- Prepare learners for the next session by previewing the upcoming topics related to crop rotation or introduction to the benefits of organic gardening.

Extended Activities:
- Activity 1: Create a garden maintenance schedule where students select a specific organic crop and outline its monthly maintenance practices.
- Activity 2: Organize a school garden day where students can apply what they've learned by working together on maintaining a designated organic gardening space.
- Activity 3: Engage in a project where students monitor an organic garden over a month, documenting their observations and discussing the successes and challenges they faced.

Teacher Self-Evaluation:

WEEK 11: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Food Production Processes
Sub-Strand: Organic Gardening

Specific Learning Outcomes:
By the end of the lesson, students should be able to:
- Select a short season crop for growing using organic gardening practices.
- Prepare a section of the school farm for growing the selected short-season crop.
- Embrace teamwork as they select the short season crop and prepare a section of the school farm for growing the crop.

Key Inquiry Question:
- What should you consider in identifying a short season crop for growing using organic gardening practices?

Learning Resources:
- Digital resources
- School farm
- Project books

Organisation of Learning

Introduction (5 minutes):
1. Briefly review key concepts from the previous lesson on organic gardening.
2. Guide students to read and discuss content from their materials that relates to the selection of short season crops, highlighting factors such as climate, soil type, and plant growth time.

Lesson Development (30 minutes):

Step 1: Research and Understand (7 minutes)
- In groups, students will discuss different factors to consider when selecting a short season crop, such as plant requirements, growth time, and suitability for organic practices. Assign each group a different aspect of crop selection to focus on.

Step 2: Crop Selection (8 minutes)
- Each group will present their findings and collaborate to choose an appropriate short season crop for the school farm. Discussion should include the pros and cons of their selected crop, using the provided resources for support.

Step 3: Preparation Planning (10 minutes)

- Once a crop is selected, groups will develop a plan for preparing a section of the school farm. This includes tasks such as soil preparation, identifying tools needed, and assigning roles for each team member.

Step 4: Implementation Preparation (5 minutes)
- Groups will outline the steps they will take to implement their plan in the following lesson, discussing any challenges they might encounter and how they will overcome them.

Conclusion (5 minutes):
1. Summarize key points covered in the lesson, reinforcing the importance of teamwork and informed decision-making in organic gardening.
2. Conduct a brief interactive Q&A session to clarify any doubts and engage students.
3. Provide a preview of the next lesson's activities involving actual planting and maintenance of the selected crop.

Extended Activities:
- Research Project: Students can choose a different short season crop and create a mini-research project that details its requirements, benefits, and uses. They will present this in class.
- Field Observation: Organize a field trip to a local organic farm where students can observe different short-season crops and organic practices in action.
- Gardening Journal: Students can maintain a gardening journal documenting the growing process of their selected crop over time, including observations, measurements, and reflections.

Teacher Self-Evaluation:

WEEK 11: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Food Production Processes
Sub Strand: Organic Gardening

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Grow a selected short-season crop using organic gardening practices.
- Carry out maintenance practices on the grown crops using organic gardening.

Key Inquiry Question(s):
- How can we produce food crops through organic gardening?

Learning Resources:
- School farm
- Short-season crops (vegetables, legumes)
- Digital devices (for research)
- Watering cans

Organisation of Learning:

Introduction (5 minutes)
- Begin with a brief review of the previous lesson on basic gardening concepts.
- Guide learners to read and discuss relevant content from the learning resources, focusing on organic gardening principles such as soil health, composting, and pest management.

Lesson Development (30 minutes)

Step 1: Selecting and Preparing the Area (10 minutes)
- In groups or pairs, learners will choose a short-season crop (e.g., lettuce, radishes, beans).
- Discuss and prepare the plot area by clearing debris and loosening the soil using hand tools. Introduce the importance of soil health for organic gardening.

Step 2: Planting the Selected Crop (10 minutes)
- Guide learners to plant the selected crop seeds in rows or designated spacing.
- Emphasize the importance of following proper planting practices such as depth and spacing, highlighting how these techniques support healthy plant growth.

Step 3: Implementing Organic Gardening Practices (5 minutes)
- Each group will discuss and decide on organic practices to use, such as using compost or natural fertilizers and pest deterrents (e.g., companion planting).

- Encourage learners to note the practices discussed for their future applications.

Step 4: Maintenance of the Crop (5 minutes)
- Discuss what maintenance practices (e.g., watering, weeding, observing for pests) are necessary after planting.
- Assign roles to each group member to ensure accountability for daily or regular maintenance over the coming weeks.

Conclusion (5 minutes)
- Summarize the key points discussed: selecting crops, planting, organic practices, and maintenance.
- Conduct a brief interactive activity such as a quiz or discussion to reinforce the main concepts, asking learners to share one organic practice they plan to implement.
- Prepare learners for the next session by previewing upcoming topics, such as pest management strategies in organic gardening.

Extended Activities:
- Home Gardening Project: Encourage students to start a small home garden using organic practices, documenting their progress, and challenges faced.
- Research Task: Assign students to research a specific type of organic pest control (e.g., using beneficial insects) and prepare a short presentation for the class.
- Field Trip: Plan a visit to a local organic farm or community garden to learn about larger-scale organic practices and their benefits.

Teacher Self-Evaluation:

WEEK 11: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	AGRICULTURE AND NUTRITION			

Strand: Food Production Processes
Sub Strand: Organic Gardening

Specific Learning Outcomes:
By the end of the lesson, learners should be able to:
- Grow a selected short-season crop using organic gardening practices.
- Carry out maintenance practices on the grown crops using organic gardening.

Key Inquiry Question(s):
- How can we produce food crops through organic gardening?

Learning Resources:
- School farm
- Short-season crops (vegetables, legumes)
- Digital devices (for research)
- Watering cans

Organisation of Learning:

Introduction (5 minutes)
- Begin with a brief review of the previous lesson on basic gardening concepts.
- Guide learners to read and discuss relevant content from the learning resources, focusing on organic gardening principles such as soil health, composting, and pest management.

Lesson Development (30 minutes)

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- Discuss what maintenance practices (e.g., watering, weeding, observing for pests) are necessary after planting.
- Assign roles to each group member to ensure accountability for daily or regular maintenance over the coming weeks.

Conclusion (5 minutes)
- Summarize the key points discussed: selecting crops, planting, organic practices, and maintenance.
- Conduct a brief interactive activity such as a quiz or discussion to reinforce the main concepts, asking learners to share one organic practice they plan to implement.
- Prepare learners for the next session by previewing upcoming topics, such as pest management strategies in organic gardening.

Extended Activities:
- Home Gardening Project: Encourage students to start a small home garden using organic practices, documenting their progress, and challenges faced.
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- Field Trip: Plan a visit to a local organic farm or community garden to learn about larger-scale organic practices and their benefits.

Teacher Self-Evaluation: